

Correction TD3

Exercice 1

1/ $i_1 = i_2$

2/ $V_e = R_1 i_1$

3/ $V_s = -R_2 i_1$

4/ $V_s = -\frac{R_2}{R_1} V_e$

$$A_V = \frac{R_2}{R_1} = 4$$

5/ Amplificateur inverseur.

Exercice 2

$$1/V^+ = \frac{\beta V_1}{1+\beta}$$

$$V^- = \frac{\alpha(V_2+V_3)+V_s}{2\alpha+1}$$

$$\frac{\alpha(V_2+V_3)+V_s}{2\alpha+1} = \frac{\beta V_1}{1+\beta}$$

$$V_s = \frac{(2\alpha+1)}{1+\beta} \beta V_1 - \alpha(V_2 + V_3)$$

$$2/V_s = \frac{(2\alpha+1)}{1+\beta} \beta V_1 - \alpha(V_2 + V_3) = \alpha(V_1 - V_2 - V_3)$$

$$\frac{(2\alpha+1)}{1+\beta} \beta V_1 = \alpha V_1$$

$$\frac{(2\alpha+1)}{1+\beta} \beta = \alpha$$

$$\beta = \frac{\alpha}{\alpha+1}$$

3/Montage soustracteur

Exercice 3

$$1/ V^+ = V_2$$

$$V^- = \frac{R_2 V_1 + R_1 V_s}{R_1 + R_2}$$

$$V_2 - \frac{R_2 V_1 + R_1 V_s}{R_1 + R_2} = \frac{V_s}{A_d}$$

$$V_s \left[\frac{1}{A_d} + \frac{R_1}{R_1 + R_2} \right] = V_2 - \frac{R_2 V_1}{R_1 + R_2}$$

$$V_s \cdot \rho \left[\frac{1}{A_d \cdot \rho} + 1 \right] = V_2 - \frac{R_2 V_1}{R_1 + R_2}$$

$$V_s = \frac{1}{1 + \frac{1}{A_d \cdot \rho}} \cdot \frac{1}{\rho} [V_2 - (1 - \rho)V_1]$$

Exercice 4

$$1/ \quad V^+ = V_1$$

$$V^- = \frac{K V_2}{1+K}$$

$$\frac{K}{1+K} V_2 = V_1$$

$$V_2 = \left(1 + \frac{1}{K}\right) V_1$$

$$2/A_v = \left(1 + \frac{1}{K}\right)$$

3/ amplificateur non inverseur

4/

$$a) \quad V^+ = V_3$$

$$V^- = \frac{K V_2 + V_s}{1 + K}$$

$$V_s = (1 + K) V_3 - K V_2$$

$$V_s = (1 + K) (V_3 - V_1)$$

b) soustracteur

Correction TD4

Exercise 1

$$1/V^+ = 0$$

$$V^- = \frac{\left(\frac{V_e}{R_1}\right) + \left(\frac{V_s}{Z_{eq}}\right)}{\left(\frac{1}{R_1}\right) + \left(\frac{1}{Z_{eq}}\right)}$$

$$Z_{eq} = \frac{R_2}{1 + jR_2C\omega}$$

$$V^+ = V^- = 0 \Rightarrow H(j\omega) = \frac{V_s}{V_e} = \frac{-\frac{R_2}{R_1}}{1 + jR_2C\omega}$$

$$2/ H_0 = -10$$

$$\omega_0 = 1000 \text{ rad/s}$$

$$3/ G(\omega_c) = \frac{G_{\max}}{\sqrt{2}}$$

$$1 + \left(\frac{\omega_c}{\omega_0}\right)^2 = 2$$

$$\omega_c = \omega_0$$

$$f_c = \frac{\omega_c}{2\pi} = 160 \text{ Hz}$$

$$3/G(\omega) \geq G(\omega_c) \rightarrow 1 + \left(\frac{\omega_c}{\omega_0}\right)^2 \leq 2 \quad \omega \leq \omega_c$$

$$\Delta\omega = \omega_c - 0 = 1000 \text{ rad/s}$$

$$4/ G_{dB}(\omega) = 20 \log(|H(j\omega)|)$$

$$G_{dB}(\omega) = 20 - 10 \log\left(1 + \left(\frac{\omega}{\omega_c}\right)^2\right)$$

$$\omega = \omega_c \rightarrow G_{dB}(\omega_c) = 17 \text{ dB}$$

$$\omega \ll \omega_c \rightarrow G_{dB}(\omega) = 20 \text{ dB}$$

$$\omega \gg \omega_c \rightarrow G_{dB}(\omega) = 20 \text{ dB} - 20 \log\left(\frac{\omega}{\omega_c}\right)$$

$$\varphi(\omega) = \arg(H(j\omega)) = \pi - \arctg\left(\frac{\omega}{\omega_c}\right)$$

$$\omega \ll \omega_c \Rightarrow \varphi(\omega) = \pi$$

$$\omega = \omega_c \Rightarrow \varphi(\omega_c) = \frac{3\pi}{4}$$

$$\omega \gg \omega_c \Rightarrow \varphi(\omega) = \frac{\pi}{2}$$

$\omega(\text{rad})$	$G_{\text{dB}}(\text{dB})$	$\rho(\text{rad})$
10	20	π
100	20	3,04
1000	17	$\frac{3\pi}{4}$
10000	0	1,67
100000	-20	$\frac{\pi}{2}$

$$v_s(t) = -5 + 8 \sin(10t + \phi_1 + \pi)$$

Exercise 2

$$1/f_c = 3\text{kHz}$$

2/

f(Hz)	300	1000	f_c	10^4	10^5
$G_{\text{dB}}(\text{dB})$	-20	-10	-3	-0,5	0

3/Filtre passe haut du premier ordre.

$$4/H(jf) = \frac{H_0 j \frac{f}{f_0}}{1 + j \frac{f}{f_0}}$$